

Building Integrated Photovoltaics (BIPV)



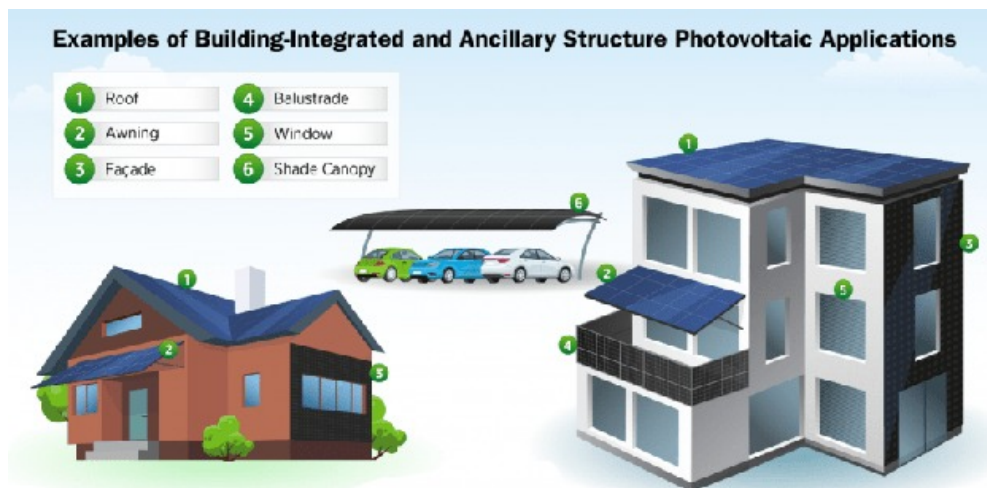
INTRODUCTION

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Photovoltaic (PV) technology is an ideal solution for the electrical supply issues that trouble the current climate-change, carbon-intensive world of power generation. PV systems can generate electricity at remote utility-operated "solar farms" or be placed directly on buildings themselves. Their fuel source is simple sunlight, and they produce electricity without the negative environmental consequences associated with other power generation methods. They are silent and reliable. The size of PV installations can range from extremely small to enormously large. They can be scaled down for small loads like

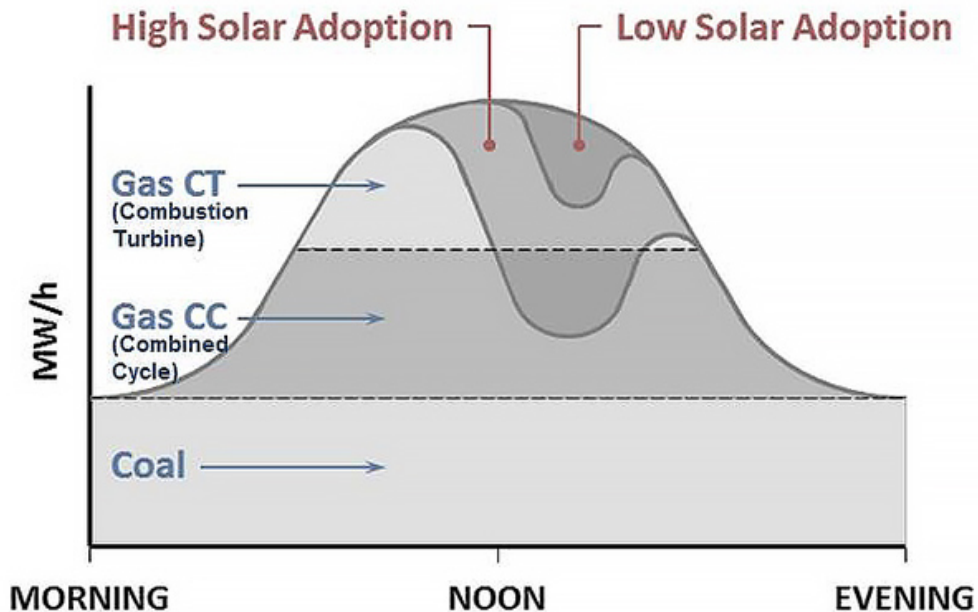
For building installations, PV systems fall into two categories, building applied photovoltaics (BAPV) and building integrated photovoltaics (BIPV). BAPV is the more common type of installation, with the solar collectors located completely outside of the building envelope. Roof-mounted, ballasted solar arrays placed on top of the roofing material are BAPV assemblies. A BIPV installation is when the photovoltaic collectors are an integral part of the building envelope. They can either replace exterior shell components or be integrated into them. Examples of BIPV components and materials currently on the market include: PV glass windows, PV glass skylights, awnings, balustrades, canopies, shingles, exterior wall panels, and even PV walkable surfaces.¹ Not only do BIPV systems generate electricity, but they can add visual interest and aesthetic design elements to the building.



Building Integrated Photovoltaics (BIPV) are when the photovoltaic collector elements are located directly within a building's envelope (or canopy structure). *Photo Credit: U.S. Department of Energy / EERE*

Building owners and utilities all benefit with the implementation of PV systems. The contribution of PV generated electricity can have major impacts on the peak demand loads that utilities have to provide power for. Late afternoon sunshine and heat accumulation in buildings lead to greater requirements placed on air conditioning systems to keep occupants cool. A building-

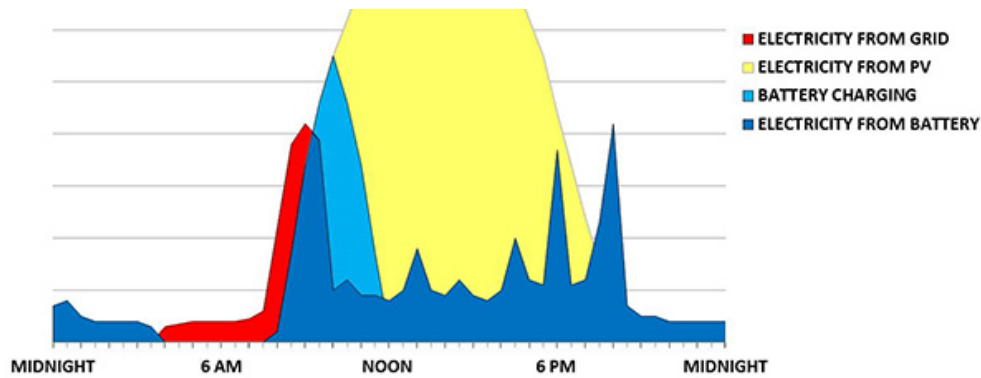
located photovoltaic system takes advantage of these same sunshine conditions to provide electricity for the building while simultaneously lessening the pressure on the utility grid to increase electricity production. The use of photovoltaics lowers the overall U.S. carbon footprint for electricity generation.



Solar energy installations have an impact on the fuel sources used by utilities to generate electricity for the grid. As PV generated power increases in the energy infrastructure, the use of higher carbon-footprint generated electricity decreases. *Image Credit: Ronald Fergle based on a graphic by Lena Hansen and Virginia Lacy of the Rocky Mountain Institute*

A building's self-consumption of the electricity generated by its PV system improves the cost-effectiveness of the installation. Buying electricity from the grid costs more than revenue achieved by selling electricity to the grid. Utilizing batteries to store PV electricity for later use can dramatically reduce the need for grid-supplied electricity. The potential for including battery storage in a PV system design should take into consideration the building loads, the time of day, the available PV generated power, and the costs for various levels of battery storage. Properly sized systems can be cost-effective for consumers.





The self-consumption of PV generated electricity coupled with battery storage can significantly reduce the need for grid-supplied electricity.

Image Credit: Ronald Fergle based on a graphic by Ralf Haselhuhn.

Depending on the fuel source, generation of electricity at a utility power plant can be inefficient and carbon-intensive, while simultaneously causing the release of Greenhouse Gases (GHGs) and harmful fine Particulate Matter (PM_{2.5}). In addition, of the electricity that enters the grid from a power plant, the U.S. Energy Information Agency (EIA) estimates that 5% is lost due to transmission and distribution (T&D) inefficiencies.²

Distributed Energy Resources (DERs) such as BIPV systems, do not have these negative environmental impacts. Solar energy is a clean, renewable energy source, and the electricity generated is already located at the point of use. For more information regarding Distributed Energy Resources, refer to the [energy.gov](https://www.energy.gov) website.

DESCRIPTION

Photovoltaic Technologies

The categories of common photovoltaic technologies used in BIPV applications include:

1. **Crystalline silicon (c-Si):** Solar cells made from solid crystalline silicon wafers (mono-crystalline or poly-crystalline/multi-crystalline) can deliver approximately 20 watts per ft² of PV array. Versions of these cells may incorporate additional layers of solar absorption materials in order to increase electrical production. Individual cells are wired together and assembled into modules at

factories before being shipped to project sites.

2. **Thin-film:** These products typically incorporate very thin layers of photovoltaic compounds that have been deposited on substrate materials using plasma enhanced, chemical vapor deposition (PECVD) processes. Commercial thin-film materials deliver about half the watts per ft² of PV array area compared to c-Si modules. Thin-film products can be rectilinear modules, rolled-out surfaces, or take the shape of an underlying architectural element. This category includes: copper indium gallium (di)selenide (CIGS), cadmium telluride (CdTe) and amorphous silicon (a-Si) cells.
3. **Emerging-PV:** These technologies include dye-sensitized solar cells (DSSC), Perovskite cells, organic cells, and quantum dot cells, among others. Efficiencies in laboratory environments range from 13% to 26%. Cells in this category can exhibit properties of transparency, flexibility, or color; and they require lower energy expenditures to create.

The DSSC cells represent a new type of solar cell that require less energy-intensive materials to manufacture, and because of their simplicity can be less costly to produce. These cells are comprised of three basic parts: the front-side glass transparent conducting oxide (TCO) electrode, an interior electrolyte solution, and a back-side counter electrode. The inside surface of the front glass is first sintered with a transparent anode, e.g., fluoride-doped tin dioxide (SnO₂:F) to make the TCO. Then it is covered with titanium dioxide (TiO₂) nanoparticles coated with photo-sensitive dyes. When the dyes are exposed to sunlight their electrons are energized and elevated into the conduction band of the TiO₂. From there they migrate to the TCO anode material. After flowing through an external circuit as electricity, the electrons re-enter the DSSC cells through a back-side counter electrode surface. The liquid electrolyte then transports the electrons back to the dye materials to re-oxidize them.

A schematic showing the movement of electrons in the DSSC photovoltaic process. *Image Credit: Ronald Fergle*

A PV installation includes:

1. **PV Modules:** These "solar collectors" can be crystalline, thin-film, or one of the emerging PV technologies. They can be transparent, semi-transparent, or opaque.
2. **Balance of System (BOS) Components:** This includes everything in a PV installation other than the solar collectors.
 - Module Mounting Systems
 - Wiring, Combiner Boxes, DC Disconnects, and AC Disconnects
 - Inverters
 - Electrical Distribution Panels
 - Batteries

PV Modules: These components are where the conversion of sunlight into electricity actually occurs. Energetic photons in sunlight excite electrons in the semi-conductor materials which elevate them to a higher energy conduction band. The electrons then become free to move as electricity within an external circuit. The electricity coming from PV modules is always Direct Current (DC).

Module Mounting Systems: BIPV mounting systems use clips, bolts, or adhesives to fix the modules directly to the envelope structure. Photovoltaic glass units for façade or roof applications are installed similarly to windows or skylights, but

with DC cabling attached. For BAPV installations these systems usually consist of metal frameworks called racking. They can be constructed to create fixed, saw-tooth arrangements or flat planes that are located close to the roofing surface. Typically weights or heavy blocks are used to secure the racking in place.

Wiring, Combiner Boxes, DC Disconnects, and AC

Disconnects: These are the components that facilitate and address the flow of electricity in the installation. Individual wiring from groups of modules can be combined into single cables in combiner boxes for circuit simplicity and to reduce the overall amount of wiring material. Combiner boxes also provide over-current protection. DC Disconnects and AC Disconnects are switches located at strategic points in the installation in order to disconnect or curtail the flow of electricity.

Inverters: These units convert the DC electricity coming from the PV modules into AC electricity. String invertors handle the output from multiple modules, and micro-inverters are dedicated to a single module.

Electrical Distribution Panels: This is the location where PV installations interconnect with a building's electrical infrastructure. Power coming from the PV system is wired into the distribution panelboard as an individual circuit. The circuit breaker on this circuit is referred to as the Over Current Protection Device (OCPD) and subject to specific sizing requirements.

Batteries: These devices store power for use at a later time. The energy flow into and out of the battery storage system is determined based on user-specified parameters or building energy management system (BMS) directives. Batteries are commonly used to store power generated from the PV array during sunny periods, and then provide that power later on to help meet the facility's energy requirements.

For more detailed information on PV module technologies and

BOS Components, refer to the related discussion on the [WBDG PV page](#).

A simplified guide for how PV modules can be connected to power optimizers, string inverters, or micro-inverters based on system design objectives. (System schematics, including combiner boxes and disconnect switches, vary based on project parameters and equipment used.) *Image Credit: Ronald Fergle*

Building Integrated Photovoltaics (BIPV) System

Building Integrated Photovoltaics is the implementation of photovoltaics as part of the building envelope. The solar collectors serve the dual function of protecting the structure from external environmental conditions, as well as being a source for electrical power. While the BIPV system itself has an initial financial cost, because it potentially replaces other building materials the overall costs of the envelope may not increase significantly. BIPV systems can also reduce HVAC electrical requirements and cooling costs when the modules are used to shade the building. When all of the advantages are taken into consideration, BIPV installations can be viewed as financial investments. They have an up-front cost, but in turn they can significantly reduce or eliminate a building's yearly energy costs, pay for themselves, and provide building owners with continuing economic savings. A recent study has documented how BIPV installations have a positive return-on-investment (ROI), and even north-facing facades can be

economically feasible.³

Design Of A Building Integrated Photovoltaics (BIPV) System

The process of designing a BIPV system is not unlike that for other building systems. Decisions should take into consideration life-cycle cost analyses in addition to up-front costs, installation procedures, performance expectations, and O&M requirements. However, with BIPV installations the aesthetics are also important and should be taken into account.

Steps in designing a BIPV system overlap, in that the consideration of one topic may impact the resolution for another. A successful solution addresses all concerns simultaneously. The general list of topics includes:

1. **Energy Conscious Building Design:** This strategy reduces overall energy use, enhances comfort, and saves money while also enabling the BIPV system to provide a greater percentage of the electricity required.
 - **Daylighting:** The use of sunlight and light from the skydome to illuminate interior building spaces. This reduces the electrical loads and heat generated from light fixtures.
 - **Thermal Mass:** Taking advantage of a material's ability to store and release heat energy in order to even out interior building temperature fluctuations.
 - **Natural Convection:** Using the natural properties of air circulation to ventilate, heat, or cool interior building spaces.
2. **Type of PV System:** Determine if the system will be grid-connected, grid-connected with battery backup, or stand-alone.
 - The majority of BIPV systems are tied to a utility grid, which in effect uses the grid as storage and backup. The system type and configuration should be

developed based on the priorities of the owner, which could include: budget limitations, space constraints, electrical requirements, energy independence, and aesthetics, among others.

- For stand-alone systems powered by PV alone, the system, including battery storage, should be sized to meet both the building's peak demand loads and the lowest power production projections of the PV array. Installations like these typically include a backup generator for unusual or excessive peak loads.

3. **Location of Installation:** Any exterior building surface is a potential location for a BIPV installation. Roof elements include: photovoltaic shingles, rolled thin-film surfaces, and PV glass skylights that have PV cells or transparent PV surfaces incorporated into them. Wall possibilities include: siding with integrated PV surfaces, PV glass windows that contain PV cells or PV coatings, and shading devices that are also PV collectors. Railings, carports, and covered entryways are additional locations. As part of the PV component selection process it is important to consider how the collector surfaces will be attached to the sub-structure. Manufacturers of PV components provide detailed information regarding mounting requirements.

4. **Building Electrical Load Analysis:** Consider the building's electrical usage patterns and adjust loads if possible to reduce peak levels. Depending on the building type (or functions occurring within the structure), shifting when power is required can reduce demand spikes and the peak loads they place on the PV system. Examples of flexible tasks include: meetings that require lighting and space conditioning, optional machinery processes, operation of dishwashers or laundry facilities, and heating of hot water for thermal storage. Electrical demands are typically greater in the afternoon because of HVAC cooling loads, so when non-time-sensitive tasks can be moved to the morning hours, the peak afternoon loads become less. Installing motion detectors on lighting systems and turning off office equipment when not in use are simple strategies

to reduce power demands. It has also been shown that educating building occupants about the benefits of reducing plug-loads helps to achieve lower energy use.⁴ In addition, it may be worthwhile to incorporate battery storage to reduce the purchase of electricity during the more expensive power demand periods.

5. **Provide Adequate Ventilation:** PV performance efficiencies are reduced by elevated operating temperatures. This affects crystalline silicon PV cells more than amorphous silicon thin-films, but all PV cells are susceptible. To improve conversion efficiency, allow appropriate ventilation behind the modules in order to dissipate heat.
6. **Consider Using PV Modules to Filter Direct Sunlight:** When using semi-transparent thin-film modules or semi-transparent crystalline modules (where the PV cells are placed apart from each other between two layers of glass), it is possible to create unique daylighting features in facades, roofing, or skylight PV systems. These elements can help reduce unwanted cooling loads and the glare associated with large expanses of architectural glazing.
7. **Incorporate PV Modules as Shading Elements:** PV arrays can double as awnings over view-glass areas of buildings and can provide appropriate shading. When sunshades are considered as part of an integrated design approach, chiller capacity can often be smaller and perimeter cooling distribution reduced or even eliminated.
8. **Design for the Local Climate and Environment:** It is important to understand the impacts of climate and environment on the array output. Cold, clear days will increase power production, while hot, overcast days will reduce array output. Typical considerations include:
 - Surfaces reflecting light onto the array (e.g., snow, lakes, or wide rivers) will increase the array output.
 - Potential snow- and wind-loading conditions may require additional bracing or structural analysis.
 - Modules angled more vertically will shed snow quicker.
 - Horizontal modules and arrays located in dry, dusty environments, or environments with heavy industrial

traffic or pollution, will require periodic rinsing with water to limit efficiency losses.

- c-Si modules have higher efficiencies and perform best in clear sky conditions, but their power output decreases significantly in cloudy or shady situations. While DSSC, CdTe, a-Si, and CIGS cell types have lower efficiencies compared to c-Si, they are less affected by cloudy or overcast conditions.

9. **Address Site Planning Issues:** Early in the design phase, ensure that the solar array will receive maximum exposure to the sun and will not be shaded by site obstructions such as nearby buildings or trees. It is important that the system be unshaded during the peak solar collection period consisting of three hours on either side of solar noon. The impact of shading on a PV array can be significant.
10. **Consider Array Orientation:** Array orientation and tilt impacts the annual energy output of a system. Arrays tilted towards the Sun generate 50%–70% more electricity than vertical façade installations, and southern facing arrays maximize power generation. However, advancements in PV technologies have increased the flexibility of array design; so it may be possible to tune the electrical output of a system to be closer to the time of day the power is required. Certain module types may be more effectively used facing east for the morning solar gain, or west for the late afternoon sunlight conditions (CdTe, CIGS, DSSC, and a-Si thin-films), and high-gain modules (typically c-Si) can be aligned slightly west of south so they produce more electricity during the afternoon peak building demand loads. As the costs for PV installations continue to decrease, the strategy to provide more continuous power generation becomes more affordable. Portions of arrays that are oriented to the east or west may not be as high in efficiency or produce the sheer volume of electricity that the southern facing portions do, but they can provide additional power closer to the time that some building loads require it.
11. **Use Credentialed Professionals:** Ensure that the designers,

installers, and maintenance professionals involved with the project are properly trained, licensed, certified, and experienced in PV systems work. They should be knowledgeable of the latest advancements in commercially available technologies, products, and installation practices.

Solar insolation levels can vary based on building surface orientation. While surfaces tilted towards the Sun receive the most energy, secondary and tertiary surfaces can still contribute meaningful amounts of PV generated electricity. *Image Credit: Ronald Fergle based on a graphic by Polysolar Ltd.*

APPLICATION

BIPV systems can be designed to blend in with traditional building materials and appearances, or they may be used to create a more innovative [aesthetic](#). The examples below show how PV modules can become attractive elements of building exteriors. Photovoltaics may be integrated into numerous assemblies within building envelopes, including:

- **Facades:** Solar cells can complement or replace traditional view windows or spandrel glass. While these installations are on vertical surfaces, which reduce the intensity of the solar insolation, the overall size of a facade can help compensate for the reduced power per unit area.
- **Awnings:** Photovoltaics may be incorporated into awnings

or slightly sloped, saw-tooth canopy designs. Semi-transparent modules provide filtered sunlight underneath while affording additional architectural benefits such as passive shading.

- **Roofing:** The use of PV in roofing systems can provide a direct replacement for batten and seam metal roofing, traditional 3-tab asphalt shingles, and ceramic tiles. Note that these types of installations require adequate ventilation in order to keep the cell temperatures cooler.
- **Skylights:** Using PV for skylight systems can be both an economical use of PV and an interesting design feature. Just as with PV windows, the semi-transparency enables visual connections to the exterior environment while providing diffuse natural lighting.

An example of the aesthetic potential of BIPV is the SwissTech Convention Center (STCC) on the Ecole Polytechnique Federale de Lausanne (EPFL) Ecublens, Switzerland, campus. The southwest façade contains 280 m² of 355 integrated Dye-Sensitized Solar Cells (DSSC), also called Grätzel cells, arranged within 65 columns of various heights. The system provides 3 kWp of electricity. The transparent DSSC installation filters direct afternoon sunlight entering the convention center main lobby; while at the same time providing a visual connection to the exterior environment with views to the sky, neighboring buildings, trees, and passersby.

Exterior views of the SwissTech Convention Center southeast façade (left) and southwest façade (right). *Photo Credit: Ronald Fergle*

Interior views of the DSSC modules exhibit a canvas of translucent vertical ribbons of color with the blue sky in the background. As the sun progresses through the sky, the colorful shadows cast from the modules move across the lobby floor. *Photo Credit: Ronald Fergle*

The light-weight modules are mounted to metal bars on the exterior side of the window glazing. The electrical cabling runs within channels next to the windows. *Photo Credit: Ronald Fergle*

Examples of c-Si wafers being used in innovative ways include the Energiewürfel building in Konstanz, Germany, and the Ludesch Community Centre in Vorarlberg, Austria. The modules have dual-glass surfaces with individual, perforated c-Si wafers spaced evenly inside. The installations filter direct sunlight while simultaneously providing views beyond. The Energiewürfel large-format, south-facing window installation has a 22% transparency, and when combined with the PV roof installation generates 23.2 kWp of electricity. The 350 m² Ludesch Community Centre canopy is comprised of 120 slightly-sloped modules oriented to the southwest, and generates 16,000 kWh/yr of electricity. The canopy emphasizes the exterior gathering area while protecting visitors from rain

and snow.

Semi-transparent module installations in the Energiewürfel building in Konstanz, Germany, (left) and the Ludesch Community Centre "town square" plaza in Vorarlberg, Austria (right). **Photo Credit: Sunways AG**

The Beit Havered building near Tel Aviv, Israel, has a photovoltaic façade composed of crystalline silicon glass with white digital printing on the surface. The printing provides a more traditional appearance while allowing the solar energy to pass through to the PV cells behind. The 608 m² installation is estimated to generate 1,938,623 kWh of electricity over 35 years, with avoided CO₂ emissions of 1,409 Tons of CO₂. The system payback period is less than 4 years.

The Beit Havered building's façade of c-Si PV glass with white digital printing on the surface provides a more traditional commercial/office building aesthetic. **Photo Credit: Onyx Solar**

The Paul Horn Arena in Tübingen, Germany, is comprised of PV modules designed to be both attractive and efficient power

generators. The aesthetics take advantage of the emerald-green "fractured" multi-crystalline silicon cell appearance mounted within oversized white rectangular frames. The unobstructed, 530 m² installation receives continuous solar insolation throughout the day. The system generates 43.7 kWp of electricity.

The south façade of the Paul Horn Arena in Tübingen, Germany. *Photo Credit: Sunways AG*

The Life Sciences Building (LSB) at the University of Washington has a 650 m² 20% transparent amorphous silicon (a-Si) vertical fin BIPV installation on the southwest curtain wall. The photovoltaic fins generate 3.15 W/ft², and over their 35 year lifespan are estimated to provide 496,885 kWh of electricity with a CO₂ avoidance of 333 Tons of CO₂.

The southwest façade of the Life Sciences Building (left) and a close-up of the semi-transparent fins (right). *Photo Credit: Onyx Solar*

The Frank Gehry designed Novartis Campus building in Basel, Switzerland, exhibits the freeform potential of BIPV. The envelope contains a combination of dual-glass PV skylights and PV window modules with imbedded, perforated PV cells. The 1,300 m² PV installation provides 92 kWp of electricity.

The Novartis Campus building southern façade (left) and outward view from the interior (right) show that photovoltaic systems do not need to dominate a building's aesthetic to be effective. *Photo Credit: Sunways AG*

RELEVANT CODES AND STANDARDS

- [*IEEE 1547 Standard for Interconnecting Distributed Resources with Electric Power Systems*](#)
- [Inflation Reduction Act of 2022 \(IRA\)](#)   and the [EPA summary](#)
- [International Code Council \(ICC\) 690.12](#)
- [NFPA 70 National Electrical Code \(NEC\) Article 690.12](#)
- [UL 1703 Standard for Flat-Plate Photovoltaic Modules and Panels](#)
- [UL 1741 Standard for Inverters, Converters, Controllers and Interconnection System Equipment for Use With Distributed Energy Resources](#)

Publications

- [Building-Integrated Photovoltaic Designs for Commercial and Institutional Structures: A Sourcebook](#)  [for Architects](#) 
- [Building-Integrated Solar Technology](#), Roland Krippner (Ed.), Detail Business Information GmbH, Munich, DE, www.detail.de, ISBN 978-3-95553-362-5 (Print), ISBN 978-3-95553-363-2 (E-Book), ISBN 978-3-95553-364-9 (Bundle)
- The SwissTech Convention Center, Richter Dahl Rocha & Associés architects SA, Editions Favre SA, Lausanne, ISBN 978-2-8289-1480-6

ADDITIONAL RESOURCES

Websites

- [Database of State Incentives for Renewables & Efficiency \(DSIRE\)](#), provides a comprehensive list of federal, state, and local incentives that promote renewable energy and energy efficiency.
- DOE's EERE [Solar Photovoltaics Technology Basics](#), gives a brief description of how the photovoltaic materials convert sunlight into electrical energy.
- [National Center for Photovoltaics \(NCPV\)](#), focuses on innovations in photovoltaic technology that drive industry growth in photovoltaic manufacturing nationwide. Formed by the U.S. Department of Energy (DOE) and based at NREL, the NCPV focuses on research and development and increasing U.S. competitiveness.
- [North American Board of Certified Energy Practitioners \(NABCEP\)](#), provides an industry certification of experienced photovoltaic installers. NABCEP was designed to raise industry standards and promote consumer confidence in photovoltaic and solar thermal system installations.
- [Procuring Solar Energy: A Guide for Federal Facility Decision Makers](#) , provides an overview for federal facility managers and their procurement terms, for the process of installing solar electric and solar thermal systems.
- [Sandia National Laboratories](#), works with the U.S. Department of Energy, industry, and academia to improve the performance and reliability of photovoltaic technologies and grid integration.

Computer-Based PV Design and Sizing Tools

- [HOMER](#)—Hybrid Optimization Model for Electric Renewables (HOMER) is a design optimization model that determines the configuration, dispatch, and load management strategy that minimizes life-cycle costs.
- NREL's [PVWatts calculator](#)—Determines the energy production and cost savings of grid-connected

photovoltaic energy systems throughout the world.

- [PV F-Chart](#)—Provides analysis and rough sizing of both grid-connected and stand-alone PV systems.
- PVFORM—Offers simulation of grid-connected and stand-alone systems, including economic analysis. Available from [Sandia National Labs](#), Albuquerque, NM.
- [TRNSYS](#)—Simulation system for renewable energy applications; originally for solar thermal, now has extensions for PV and wind.

Other

- [Solar-Estimate.org](#) is a free public service offering solar estimating tools and is supported by the Department of Energy and the California Energy Commission.

Training Courses

- [Solar energy courses in WBDG continuing education](#)

Endnotes

¹ [Onyx Solar](#), Products and Services. [↑](#)

² U.S. Energy Information Administration, [Frequently Asked Questions \(FAQS\)](#). [↑](#)

³ "[Economic analysis of BIPV systems as a building envelope material for building skins in Europe](#)", by Hassan Gholami and Harald Nils Røstvik; Department of Safety, Economics and Planning, University of Stavanger, Kjell Arholmsgate 41, 4036, Stavanger, Norway. [↑](#)

⁴ "

[Sustainability in Practice, Building and Running 343 Second Street, The Packard Foundation Headquarters](#)



", by Robert H. Knapp, Physics and Sustainable Design, Evergreen State College. [↑](#)

WBDG PARTICIPATING AGENCIES